

Practice Quiz 10D — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 10

Images and inverse images; the geometric sum formula.

1. (a) Let $A = \{-1, 0, 1, 2\}$, $B = \{a, b, c\}$, and define

$$f(-1) = a, \quad f(0) = a, \quad f(1) = b, \quad f(2) = c.$$

For $X = \{0, 2\}$, compute $f(X)$ and $f^{-1}(f(X))$. Are the latter and X equal?

Solution $f(X) = \{a, c\}$. The elements mapped into $\{a, c\}$ are $-1, 0, 2$, so

$$f^{-1}(f(X)) = \{-1, 0, 2\} \neq \{0, 2\} = X.$$

The extra element -1 appears because $f(-1) = f(0) = a$. ■

- (b) For real $r \neq 1$, state and prove by induction a closed formula for

$$r^2 + r^3 + \cdots + r^{n+2} \quad (n \geq 0).$$

Solution The formula is

$$\sum_{i=2}^{n+2} r^i = r^2 \frac{r^{n+1} - 1}{r - 1}.$$

For $n = 0$, both sides equal r^2 . Suppose the formula holds for n . Then

$$\begin{aligned} \sum_{i=2}^{n+3} r^i &= r^2 \frac{r^{n+1} - 1}{r - 1} + r^{n+3} \\ &= \frac{r^2(r^{n+1} - 1) + r^{n+3}(r - 1)}{r - 1} \\ &= \frac{r^{n+4} - r^2}{r - 1} = r^2 \frac{r^{n+2} - 1}{r - 1}. \end{aligned}$$

This is the formula for $n + 1$, so induction proves the result. ■